

LAMP LLM to ASP Modeling Protocol

Task

ASP rule induction. Given facts F and a target answer set A^* , find a small program Π with

$$AS(F \cup \Pi) = \{A^*\}.$$

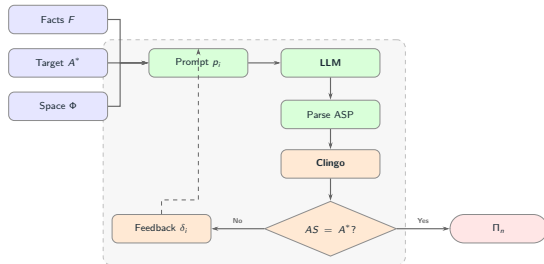
Bounded unary normal rules.

Loop

- LLM proposes rules from F , A^* , schema bounds.
- CLINGO verifies the stable model.
- Mismatches feed back. Up to 3 attempts.

Setup

120 synthetic benchmarks varying rule count, size, and negation. ILASP baseline runs search using an ASP meta encoding with 7 minute timeout.



Results

System	1st	Final	Fails
<i>gpt-5-mini</i>	90.8%	99.2%	1
<i>gpt-oss</i>	83.3%	98.3%	2
<i>qwen3-coder</i>	19.2%	64.2%	43
ILASP	—	97.5%	3

Outputs averaged 5 literals vs 22 or 33, more stratified.
Strong LAMP models solved all 3 ILASP timeout benchmarks.